

8.1 - 4

Find the eigenvalues. Find the corresponding eigenvectors.

Use the given λ or factor in Probs. 11 and 15.

$$4. \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Let's define its characteristic polynomial

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad A - \lambda I = \begin{bmatrix} 1-\lambda & 2 \\ 2 & 4-\lambda \end{bmatrix}$$

$$\begin{aligned} \det(A - \lambda I) &= (1-\lambda)(4-\lambda) - 4 = 0 \\ &= \cancel{4} - \lambda - \cancel{4\lambda} + \lambda^2 - \cancel{4} = 0 \\ &= \lambda^2 - 5\lambda = 0 \end{aligned}$$

then

$$\lambda^2 - 5\lambda = 0$$

$$\lambda(\lambda - 5) = 0$$

$$\lambda = 0 \quad \wedge \quad \lambda = 5$$

Now the eigen vectors

For $\lambda = 0$

$$A - \lambda I = A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$x_1 + 2x_2 = 0$$

$$x_1 = -2x_2$$

$$v_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

For $\lambda = 5$

$$A - \lambda I = A - 5I = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix}$$

$$-4x_1 + 2x_2 = 0$$

$$x_2 = 2x_1$$

$$v_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

8.1 - 11

Find the eigenvalues. Find the corresponding eigenvectors.

Use the given λ or factor in Probs. 11 and 15.

11. $\begin{bmatrix} 6 & 2 & -2 \\ 2 & 5 & 0 \\ -2 & 0 & 7 \end{bmatrix}, \lambda = 3$

Let's define its characteristic polynomial:

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 6-\lambda & 2 & -2 \\ 2 & 5-\lambda & 0 \\ -2 & 0 & 7-\lambda \end{vmatrix} = (7-\lambda)[(6-\lambda)(5-\lambda)-4] - 2[2(5-\lambda)] \\ &= (7-\lambda)(6-\lambda)(5-\lambda) - 4(7-\lambda) - 4(5-\lambda) \\ &= (42 - 13\lambda + \lambda^2)(5-\lambda) - 28 + 4\lambda - 20 + 4\lambda \\ &= 210 - 42\lambda - 65\lambda + 13\lambda^2 + 5\lambda^2 - \lambda^3 - 48 + 8\lambda \\ &= -\lambda^3 + 18\lambda^2 - 99\lambda + 162 \end{aligned}$$

because $\lambda = 3$ is a solution, let's divide by $\lambda - 3$

$$\begin{array}{r|rrrr} 3 & -1 & 18 & -99 & 162 \\ & & -3 & 45 & -162 \\ \hline & -1 & 15 & -54 & 0 \end{array}$$

then $\det(A - \lambda I) = (\lambda - 3)(-\lambda^2 + 15\lambda - 54)$

Multiplying by -1 and finding remaining factors

$$\det(A - \lambda I) = (\lambda - 3)(\lambda^2 - 15\lambda + 54) = 0$$

$$(\lambda - 3)(\lambda - 9)(\lambda - 6) = 0$$

then the eigen values are:

$$\lambda_1 = 3$$

$$\lambda_2 = 6$$

$$\lambda_3 = 9$$

$$\begin{array}{r} 54 \\ 27 \\ 9 \\ 3 \end{array} \left| \begin{array}{r} 16 \\ 16 \\ 16 \\ 16 \end{array} \right. \begin{array}{r} 16 \\ 16 \\ 16 \\ 16 \end{array}$$

Finally, compute the eigen vectors

$$\lambda_1 = 3$$

$$(A - 3I)x = 0$$

$$\begin{bmatrix} 6-3 & 2 & -2 \\ 2 & 5-3 & 0 \\ -2 & 0 & 7-3 \end{bmatrix} = \begin{bmatrix} 3 & 2 & -2 \\ 2 & 2 & 0 \\ -2 & 0 & 4 \end{bmatrix} \xrightarrow{\frac{1}{3}R_1 \rightarrow R_1} \begin{bmatrix} 1 & \frac{2}{3} & -\frac{2}{3} \\ 2 & 2 & 0 \\ -2 & 0 & 4 \end{bmatrix} \xrightarrow{\begin{array}{l} R_2 - 2R_1 \rightarrow R_2 \\ R_3 + 2R_1 \rightarrow R_3 \end{array}}$$

$$\begin{bmatrix} 1 & \frac{2}{3} & -\frac{2}{3} \\ 0 & \frac{2}{3} & \frac{4}{3} \\ 0 & \frac{4}{3} & \frac{8}{3} \end{bmatrix} \xrightarrow{\frac{3}{2}R_2 \rightarrow R_2} \begin{bmatrix} 1 & \frac{2}{3} & -\frac{2}{3} \\ 0 & 1 & \frac{2}{3} \\ 0 & \frac{4}{3} & \frac{8}{3} \end{bmatrix} \xrightarrow{\begin{array}{l} R_1 - \frac{2}{3}R_2 \rightarrow R_1 \\ R_3 - \frac{4}{3}R_2 \rightarrow R_3 \end{array}} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & \frac{2}{3} \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_2 + 2x_3 = 0$$

$$x_2 = -2x_3$$

$$x_1 - 2x_3 = 0$$

$$x_1 = 2x_3$$

$$x_3 = 1 \text{ then } V_1 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

$$\lambda = 6$$

$$(A - 6I)x = 0$$

$$\begin{bmatrix} 0 & 2 & -2 \\ 2 & -1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 + 2R_3 \rightarrow R_1} \begin{bmatrix} -4 & 2 & 0 \\ 2 & -1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \xrightarrow{-R_2 \rightarrow R_2} \begin{bmatrix} -4 & 2 & 0 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 - 2R_2 \rightarrow R_1}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \text{ then: } \begin{array}{l} -2x_1 + x_2 = 0 \\ -2x_1 + x_3 = 0 \end{array} \Rightarrow \begin{array}{l} x_2 = 2x_1 \\ x_3 = 2x_1 \end{array} \quad x_1 = 1$$

$$V_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$\lambda = 9$$

$$(A - 9I)x = 0$$

$$\begin{bmatrix} -3 & 2 & -2 \\ 2 & -4 & 0 \\ -2 & 0 & -2 \end{bmatrix} \xrightarrow{R_1 + R_2 \rightarrow R_1} \begin{bmatrix} -1 & -2 & -2 \\ 2 & -4 & 0 \\ -2 & 0 & -2 \end{bmatrix} \xrightarrow{-R_1 \rightarrow R_1} \begin{bmatrix} 1 & 2 & 2 \\ 2 & -4 & 0 \\ -2 & 0 & -2 \end{bmatrix} \xrightarrow{\begin{array}{l} R_2 - 2R_1 \rightarrow R_2 \\ R_3 + 2R_1 \rightarrow R_3 \end{array}}$$

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & -8 & -4 \\ 0 & 4 & 2 \end{bmatrix} \xrightarrow{R_2 + 2R_3 \rightarrow R_2} \begin{bmatrix} 1 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 4 & 2 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 2 \\ 0 & 4 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\frac{1}{2}R_2 \rightarrow R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{then: } \begin{array}{l} x_1 + x_3 = 0 \\ 2x_2 + x_3 = 0 \end{array} \Rightarrow \begin{array}{l} x_1 = -x_3 \\ x_2 = -\frac{1}{2}x_3 \end{array} \quad x_3 = 2$$

$$V_3 = \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$$

8.1-12

Find the eigenvalues. Find the corresponding eigenvectors.
Use the given λ or factor in Probs. 11 and 15.

$$12. \begin{bmatrix} 3 & 5 & 3 \\ 0 & 4 & 6 \\ 0 & 0 & 1 \end{bmatrix}$$

Let's define its characteristic polynomial

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 5 & 3 \\ 0 & 4-\lambda & 6 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)(3-\lambda)(4-\lambda)$$

then: $\lambda_1 = 1$

$\lambda_2 = 3$

$\lambda_3 = 4$

Let's compute the eigen vectors

$\lambda_1 = 1$

$$(A - I)x = 0$$

$$\begin{bmatrix} 2 & 5 & 3 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\frac{1}{3}R_2 \rightarrow R_2} \begin{bmatrix} 2 & 5 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 - 5R_2 \rightarrow R_1} \begin{bmatrix} 2 & 0 & -7 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

then $2x_1 - 7x_3 = 0 \Rightarrow x_1 = \frac{7}{2}x_3$
 $x_2 + 2x_3 = 0 \Rightarrow x_2 = -2x_3$ $x_3 = 2$

$v_1 = \begin{bmatrix} 7 \\ -4 \\ 2 \end{bmatrix}$

$\lambda_2 = 3$

$$(A - 3I)x = 0$$

$$\begin{bmatrix} 0 & 5 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -2 \end{bmatrix} \xrightarrow{-\frac{1}{2}R_3 \rightarrow R_3} \begin{bmatrix} 0 & 5 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} 5x_2 + 3x_3 &= 0 \\ x_2 + 6x_3 &= 0 \\ x_3 &= 0 \end{aligned} \Rightarrow x_2 = x_3 = 0$$

$v_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$\lambda_3 = 4$

$$(A - 4I)x = 0$$

$$\begin{bmatrix} -1 & 5 & 3 \\ 0 & 0 & 6 \\ 0 & 0 & -3 \end{bmatrix} \xrightarrow{\begin{matrix} R_3 + \frac{1}{2}R_2 \rightarrow R_3 \\ \frac{1}{6}R_2 \rightarrow R_2 \end{matrix}} \begin{bmatrix} -1 & 5 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 - 3R_2 \rightarrow R_1} \begin{bmatrix} -1 & 5 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$-x_1 + 5x_2 = 0$$

$$x_3 = 0$$

$x_1 = 5x_2$
 $v_3 = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}$

8.1-24

24. Inverse matrix. Show that A^{-1} exists if and only if the eigenvalues $\lambda_1, \dots, \lambda_n$ are all nonzero, and then A^{-1} has the eigenvalues $1/\lambda_1, \dots, 1/\lambda_n$.

Let's divide this in two parts: 1) show that A^{-1} exists
2) show that $\frac{1}{\lambda_i}$ are A^{-1} Eigen Values

1) Show that A^{-1} exists

A^{-1} exists iff $\det(A) \neq 0$

λ_i are the solutions of the characteristic polynomial

$$\det(A - \lambda I) = 0$$

If exists a $\lambda_i = 0$

$$\det(A - \lambda_i I) = 0$$

$$\det(A - 0I) = 0$$

$$\det(A) = 0$$

Then if $\lambda_i = 0$ A^{-1} does not exist
then A^{-1} exists if $\lambda_i \neq 0$

2) A^{-1} has eigen values $\frac{1}{\lambda_i}$

By definition of eigen vectors

$$A v_i = \lambda_i v_i$$

for v_i eigen vectors

A^{-1} exists then:

$$A^{-1} A v_i = A^{-1} \lambda_i v_i$$

$$v_i = A^{-1} \lambda_i v_i$$

$$\frac{1}{\lambda_i} v_i = A^{-1} v_i$$

then the eigen values of A^{-1} are the inverses of λ_i

8.3 - 3

Are the following matrices symmetric, skew-symmetric, or orthogonal? Find the spectrum of each, thereby illustrating Theorems 1 and 5. Show your work in detail.

3. $\begin{bmatrix} 2 & 8 \\ -8 & 2 \end{bmatrix}$

- It is NOT Symmetric because $C_{12} \neq C_{21}$
- It is NOT skew-symmetric because $C_{11} \neq 0$

$$A^T A \stackrel{?}{=} I$$

$$\begin{bmatrix} 2 & 8 \\ 8 & 2 \end{bmatrix} \begin{bmatrix} 2 & 8 \\ -8 & 2 \end{bmatrix} = \begin{bmatrix} 68 & 0 \\ 0 & 68 \end{bmatrix} \quad \text{It is not orthogonal}$$

Let's compute the characteristic polynomial

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 2-\lambda & 8 \\ -8 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 + 64 \\ &= (2-\lambda+8i)(2-\lambda-8i) \\ 2-\lambda+8i &= 0 & 2-\lambda-8i &= 0 \\ 2+8i &= \lambda & 2-8i &= \lambda \end{aligned}$$

then

Spectrum: $2 \pm 8i$

8.3 - 6

Are the following matrices symmetric, skew-symmetric, or orthogonal? Find the spectrum of each, thereby illustrating Theorems 1 and 5. Show your work in detail.

6.
$$\begin{bmatrix} a & k & k \\ k & a & k \\ k & k & a \end{bmatrix}$$

• It IS symmetric as $C_{ij} = C_{ji}$

• It is NOT skew-symmetric because $C_{12} = C_{21}$

It would be skew-symmetric if $a = k = 0$

Lets compute $A^T A \stackrel{?}{=} I$

$$\begin{bmatrix} a & k & k \\ k & a & k \\ k & k & a \end{bmatrix} \begin{bmatrix} a & k & k \\ k & a & k \\ k & k & a \end{bmatrix} = \begin{bmatrix} a^2 + 2k^2 & 2ak + k^2 & 2ak + k^2 \\ 2ak + k^2 & a^2 + 2k^2 & 2ak + k^2 \\ 2ak + k^2 & 2ak + k^2 & a^2 + 2k^2 \end{bmatrix}$$

then A is orthogonal if

$$a^2 + 2k^2 = 1 \quad \wedge \quad 2ak + k^2 = 0$$

Lets define its Characteristic Polynomial

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} a-\lambda & k & k \\ k & a-\lambda & k \\ k & k & a-\lambda \end{vmatrix} = (a-\lambda) [(a-\lambda)(a-\lambda) - k^2] - k [k(a-\lambda) - k^2] \\ &\quad + k [k^2 - k(a-\lambda)] \\ &= (a-\lambda)^3 - k^2(a-\lambda) - k^2(a-\lambda) + \underline{k^3} + \underline{k^3} - k^2(a-\lambda) \\ &= (a-\lambda)^3 - 3k^2(a-\lambda) + 2k^3 \\ &= (a-\lambda-k)(a-\lambda)^2 + k(a-\lambda) - 2k^2 \\ &= (a-\lambda-k)^2(a-\lambda+2k) \end{aligned}$$

then

$$(a-\lambda-k)^2 = 0 \quad \wedge \quad (a-\lambda+2k) = 0$$

$$\lambda = a-k$$

$$\lambda = 2k+a$$

Spectrum: $a-k$
 $2k+a$

8.3 -8

Are the following matrices symmetric, skew-symmetric, or orthogonal? Find the spectrum of each, thereby illustrating Theorems 1 and 5. Show your work in detail.

$$8. \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

• It is not symmetric because $C_{32} \neq C_{23}$
It would be symmetric if $\theta = 0$

• It is not skew-symmetric because $C_{11} \neq 0$

Let's compute

$$A^T A \stackrel{?}{=} I$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos^2 \theta + \sin^2 \theta & 0 \\ 0 & 0 & \sin^2 \theta + \cos^2 \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

It is orthogonal

Let's compute its characteristic polynomial

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & \cos \theta - \lambda & -\sin \theta \\ 0 & \sin \theta & \cos \theta - \lambda \end{vmatrix} = (1-\lambda) [(\cos \theta - \lambda)^2 + \sin^2 \theta]$$

then

$$1-\lambda=0 \quad \vee \quad (\cos \theta - \lambda)^2 + \sin^2 \theta = 0$$

$$\lambda = 1 \quad \cos^2 \theta - 2\lambda \cos \theta + \lambda^2 + \sin^2 \theta = 0$$

$$\lambda^2 - 2\lambda \cos \theta + 1 = 0$$

$$\lambda = \frac{2 \cos \theta \pm \sqrt{4 \cos^2 \theta - 4}}{2}$$

$$\lambda = \cos \theta \pm \sqrt{\cos^2 \theta - 1}$$

$$\lambda = \cos \theta \pm i \sin \theta$$

Spectrum

$$\lambda = 1$$

$$\lambda = \cos \theta + i \sin \theta$$

$$\lambda = \cos \theta - i \sin \theta$$

$$|\lambda| = 1$$

8.5 - 1

Is the given matrix Hermitian? Skew-Hermitian? Unitary?

Find its eigenvalues and eigenvectors.

$$1. \begin{bmatrix} 6 & i \\ -i & 6 \end{bmatrix}$$

$$\bar{A}^T = \begin{bmatrix} 6 & i \\ -i & 6 \end{bmatrix} = A \neq -A \text{ it is is Hermitian}$$

it is not Hermitian

$$\bar{A}^T A = \begin{bmatrix} 6 & i \\ -i & 6 \end{bmatrix} \begin{bmatrix} 6 & i \\ -i & 6 \end{bmatrix} = \begin{bmatrix} 37 & -12i \\ -12i & 37 \end{bmatrix} \text{ it is not Unitary}$$

$$\det(A - \lambda I) = \begin{vmatrix} 6-\lambda & i \\ -i & 6-\lambda \end{vmatrix} = (6-\lambda)^2 - 1$$

Eigen Values
 $\lambda_1 = 5$
 $\lambda_2 = 7$

$$= (6-\lambda+1)(6-\lambda-1) = (7-\lambda)(5-\lambda)$$

$$\lambda_1 = 5$$

$$(A - 5I)x = 0$$

$$\begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} \xrightarrow{R_2 + iR_1 \rightarrow R_2} \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix} \text{ then } x_1 + i x_2 = 0$$

$$x_1 = -i x_2$$

$$x_2 = 1$$

$$v_1 = \begin{bmatrix} -i \\ 1 \end{bmatrix}$$

$$\lambda_2 = 7$$

$$(A - 7I)x = 0$$

$$\begin{bmatrix} -1 & i \\ -i & -1 \end{bmatrix} \xrightarrow{-R_1 \rightarrow R_1} \begin{bmatrix} 1 & -i \\ -i & -1 \end{bmatrix} \xrightarrow{R_2 + iR_1 \rightarrow R_2} \begin{bmatrix} 1 & -i \\ 0 & 0 \end{bmatrix}$$

$$\text{then } x_1 - i x_2 = 0$$

$$x_1 = i x_2$$

$$x_2 = 1$$

$$v_2 = \begin{bmatrix} i \\ 1 \end{bmatrix}$$

8.5 - 2

Is the given matrix Hermitian? Skew-Hermitian? Unitary?

Find its eigenvalues and eigenvectors.

$$2. \begin{bmatrix} i & 1+i \\ -1+i & 0 \end{bmatrix}$$

$$\bar{A}^T = \begin{bmatrix} i & -1-i \\ 1-i & 0 \end{bmatrix} = -A \neq A \quad \begin{array}{l} \text{It is not Hermitian} \\ \text{It is skew Hermitian} \end{array}$$

$$\bar{A}^T A = \begin{bmatrix} i & -1-i \\ 1-i & 0 \end{bmatrix} \begin{bmatrix} i & 1+i \\ -1+i & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1-i \\ 1+i & 2 \end{bmatrix} \quad \text{It is not Unitary}$$

$$\det(A - \lambda I) = \begin{vmatrix} i-\lambda & 1+i \\ -1+i & -\lambda \end{vmatrix} = -\lambda(-\lambda+i) - (1+i)(-1+i)^2$$

$$= \lambda^2 - i\lambda + 2$$

$$= (\lambda - 2i)(\lambda + i)$$

$$\lambda_1 = 2i$$

$$\lambda_2 = -i$$

then the eigen vectors

$$\lambda_1 = 2i$$

$$(A - 2iI)x = 0$$

$$\begin{bmatrix} -i & -1+i \\ -1+i & -2i \end{bmatrix} \xrightarrow{iR_1 \rightarrow R_1} \begin{bmatrix} 1 & 1-i \\ -1+i & -2i \end{bmatrix} \xrightarrow{R_2 + (1-i)R_1 \rightarrow R_2} \begin{bmatrix} 1 & -1+i \\ 0 & 0 \end{bmatrix}$$

$$\text{then } x_1 + (-1+i)x_2 = 0$$

$$\Downarrow$$

$$x_1 = 1-i x_2$$

$$x_2 = 1$$

$$V_1 = \begin{bmatrix} 1-i \\ 1 \end{bmatrix}$$

$$\lambda_2 = -i$$

$$\begin{bmatrix} 2i & 1+i \\ -1+i & i \end{bmatrix} \xrightarrow{-\frac{i}{2}R_1 \rightarrow R_1} \begin{bmatrix} 1 & \frac{1-i}{2} \\ -1+i & i \end{bmatrix} \xrightarrow{R_2 + (1-i)R_1 \rightarrow R_2} \begin{bmatrix} 1 & \frac{1-i}{2} \\ 0 & 0 \end{bmatrix}$$

$$x_1 + \frac{1-i}{2}x_2 = 0$$

$$x_1 = \frac{i-1}{2}x_2$$

$$x_2 = 2$$

$$V_2 = \begin{bmatrix} i-1 \\ 2 \end{bmatrix}$$

8.5 - 5

Is the given matrix Hermitian? Skew-Hermitian? Unitary?

Find its eigenvalues and eigenvectors.

$$5. \begin{bmatrix} i & 0 & 0 \\ 0 & 0 & i \\ 0 & i & 0 \end{bmatrix}$$

$$\overline{A}^T = \begin{bmatrix} -i & 0 & 0 \\ 0 & 0 & -i \\ 0 & -i & 0 \end{bmatrix} = -A \neq A \quad \begin{array}{l} \text{It is not Hermitian} \\ \text{It is skew-Hermitian} \end{array}$$

$$\overline{A}^T A = \begin{bmatrix} -i & 0 & 0 \\ 0 & 0 & -i \\ 0 & -i & 0 \end{bmatrix} \begin{bmatrix} i & 0 & 0 \\ 0 & 0 & i \\ 0 & i & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{It is Unitary}$$

$$\det(A - \lambda I) = \begin{vmatrix} i-\lambda & 0 & 0 \\ 0 & -\lambda & i \\ 0 & i & -\lambda \end{vmatrix} = (i-\lambda)(\lambda^2 + 1) \\ = (i-\lambda)(\lambda+i)(\lambda-i) \\ \lambda = i \vee \lambda = -i \vee \lambda = i$$

Eigen Values

$$\lambda_1 = i \\ \lambda_2 = -i$$

Lets compute Eigen Vectors

$$\lambda_1 = i$$

$$(A - iI)x = 0$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & -i & i \\ 0 & i & -i \end{bmatrix} \xrightarrow{iR_2 \rightarrow R_2, iR_3 \rightarrow R_3} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \xrightarrow{R_3 + R_2 \rightarrow R_3} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{then} \\ x_2 = x_3 \\ x_1 \end{array}$$

$$V_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda_2 = -i$$

$$(A + iI)x = 0$$

$$\begin{bmatrix} 2i & 0 & 0 \\ 0 & i & i \\ 0 & i & i \end{bmatrix} \xrightarrow{\frac{1}{2i}R_1 \rightarrow R_1, R_3 + R_2 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & i & i \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\frac{1}{i}R_2 \rightarrow R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{then } x_1 = 0 \\ x_2 = -x_3 \end{array}$$

$$V_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

8.5 - 7

7. **Pauli spin matrices.** Find the eigenvalues and eigenvectors of the so-called *Pauli spin matrices* and show that $S_x S_y = iS_z$, $S_y S_x = -iS_z$, $S_x^2 = S_y^2 = S_z^2 = \mathbf{I}$, where

$$S_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad S_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix},$$

$$S_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

For S_x

$$\det(S_x - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - 1 = (\lambda + 1)(\lambda - 1) \quad \begin{matrix} \lambda_1 = 1 \\ \lambda_2 = -1 \end{matrix}$$

$$\lambda_1 = 1$$

$$(S_x - I)x = 0$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \xrightarrow{R_1 + R_2 \rightarrow R_2} \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{matrix} \text{then} \\ -x_1 + x_2 = 0 \\ x_2 = x_1 \end{matrix} \quad v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = -1$$

$$(S_x + I)x = 0$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \xrightarrow{R_2 - R_1 \rightarrow R_2} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{matrix} \text{then} \\ x_1 + x_2 = 0 \\ x_1 = -x_2 \end{matrix} \quad v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

For S_y

$$\det(S_y - \lambda I) = \begin{vmatrix} -\lambda & -i \\ i & -\lambda \end{vmatrix} = \lambda^2 - 1 = (\lambda + 1)(\lambda - 1) \quad \begin{matrix} \lambda_1 = 1 \\ \lambda_2 = -1 \end{matrix}$$

$$\lambda_1 = 1$$

$$\begin{bmatrix} -1 & -i \\ i & -1 \end{bmatrix} \xrightarrow{-R_1 \rightarrow R_1} \begin{bmatrix} 1 & i \\ i & -1 \end{bmatrix} \xrightarrow{R_2 - iR_1 \rightarrow R_2} \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix} \quad \begin{matrix} \text{then} \\ x_1 = -i x_2 \end{matrix} \quad v_1 = \begin{bmatrix} -i \\ 1 \end{bmatrix}$$

$$\lambda_2 = -1$$

$$\begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} \xrightarrow{R_2 + iR_1 \rightarrow R_2} \begin{bmatrix} 1 & -i \\ 0 & 0 \end{bmatrix} \quad \begin{matrix} \text{then} \\ x_1 = i x_2 \end{matrix} \quad v_2 = \begin{bmatrix} i \\ 1 \end{bmatrix}$$

For S_z

$$\det(S_z - \lambda I) = \begin{vmatrix} 1-\lambda & 0 \\ 0 & -1-\lambda \end{vmatrix} = (1-\lambda)(-1-\lambda) = \lambda^2 + \lambda - 1$$
$$= \lambda^2 - 1 \quad \lambda_1 = 1$$
$$= (\lambda+1)(\lambda-1) \quad \lambda_2 = -1$$

$\lambda_1 = 1$

$$(S_z - I)x = 0$$

$$\begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix}$$

then $-2x_2 = 0$
 $x_2 = 0$ $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\lambda_2 = -1$

$$\begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix}$$

then $-2x_1 = 0$
 $x_1 = 0$ $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Now

$$S_x S_y = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = i S_z$$

$$S_y S_x = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} = -i S_z$$

$$S_x^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$S_y^2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$S_z^2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

8.5 - 13

13. Product. Show that $\overline{(ABC)}^T = -C^{-1}BA$ for any $n \times n$ Hermitian A , skew-Hermitian B , and unitary C .

By definition

$$A = \bar{A}^T$$

$$\bar{B}^T = -B$$

$$\bar{C}^T C = I \Rightarrow \bar{C}^T = C^{-1}$$

then:

$$\begin{aligned} \overline{(ABC)}^T &= (\bar{A} \bar{B} \bar{C})^T \\ &= \bar{C}^T \bar{B}^T \bar{A}^T \\ &= \bar{C}^{-1} - B \cdot A \\ &= -C^{-1} B A \end{aligned}$$